GLPI + LibreNMS

Ghid complet de instalare, configurare si operare pentru Cisco si MikroTik

anatolie-ernu/GLPI-LibreNMS

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1 Despre acest document

Acest ghid reuneste si extinde documentatia repository-ului **GLPI-LibreNMS** intr-un document operational unic. Scopul este implementarea unei platforme self-hosted care poate raspunde la doua intrebari diferite:

1. **Ce este acest asset?** - GLPI.
2. **Unde este conectat in retea?** - LibreNMS.

Rezultatul urmarit este corelarea:

```
Computer -> IP -> MAC -> VLAN -> Switch -> Physical Port
```

Stack-ul este destinat in principal retelelor enterprise cu switch-uri **Cisco IOS/IOS XE** si **MikroTik RouterOS**.

> Important: un MAC invatat pe un uplink/trunk nu demonstreaza ca endpoint-ul este conectat fizic pe acel port. Pentru localizare se folosesc impreuna FDB/MAC, VLAN, LLDP/CDP si topologia downstream.

2 1. Componente

Componenta	Rol
GLPI	Asset Management / CMDB / inventar hardware si software
GLPI Agent	Inventarierea endpoint-urilor Windows/Linux
LibreNMS	Network discovery, SNMP polling, FDB/MAC, VLAN, LLDP/CDP
MySQL	Baza de date GLPI
MariaDB	Baza de date LibreNMS
Redis	Cache si sesiuni LibreNMS
LibreNMS Dispatcher	Polling/discovery sidecar

2.1 1.1 Ownership-ul datelor

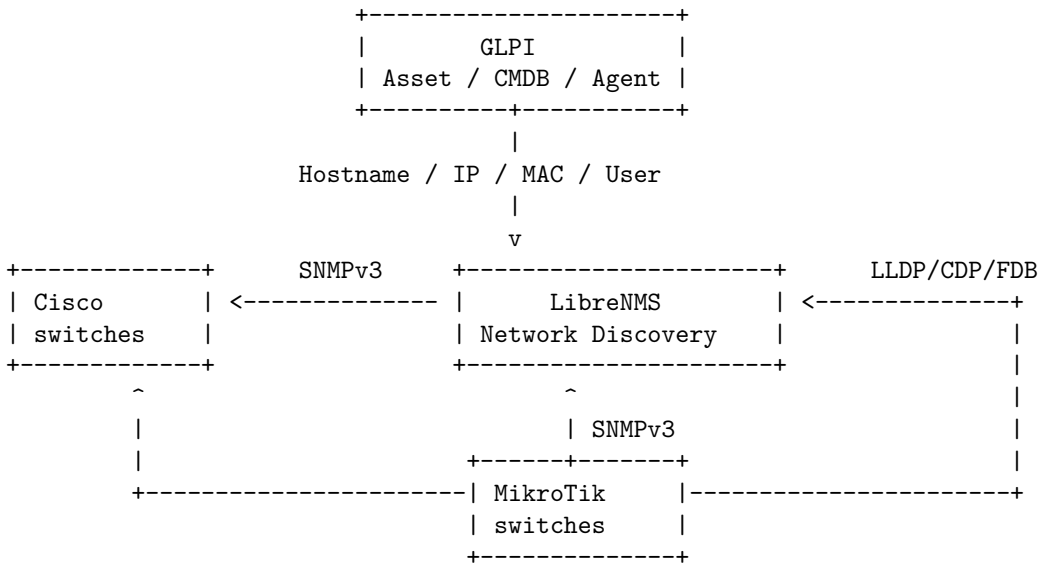
GLPI este autoritativ pentru:

- calculatoare si servere;
- serial number, vendor, model;
- CPU, RAM, storage;
- OS si software instalat;
- user/departament/locatie;
- lifecycle si asset metadata;
- IP/MAC raportate de GLPI Agent.

LibreNMS este autoritativ pentru:

- starea curenta a switch-urilor/routerelor;
- interfaces si counters;
- VLAN-uri;
- FDB/MAC tables;
- ARP/NDP unde sunt disponibile;
- LLDP/CDP neighbors;
- port utilization/errors;
- device availability si sensors.

3 2. Arhitectura



Docker foloseste retele separate `glpi-net` si `librenms-net`. In Stage 1 nu este necesara comunicarea directa GLPI-LibreNMS. O integrare API poate fi adaugata ulterior ca serviciu separat.

4 3. Cerinte recomandate

Resursa	Baseline recomandat
OS	Ubuntu Server 24.04 LTS
CPU	4-8 vCPU
RAM	16 GB
Storage	200-250 GB SSD minim
Network	IP static in management LAN
DNS/NTP	Configurate si functionale
Backup	Storage separat de host

LibreNMS poate genera o cantitate semnificativa de RRD/history data. Monitorizeaza permanent capacitatea filesystem-ului Docker.

5 4. Cerinte firewall si routing

Flux	Port	Directie	Status
SNMP polling	UDP/161	LibreNMS -> device	Obligatoriu
ICMP	ICMP	LibreNMS -> device	Recomandat
GLPI web	TCP/8081 implicit	Admin/Agent -> host	Stage 1
LibreNMS web	TCP/8000 implicit	Admin -> host	Stage 1
SNMP traps	UDP/162	Device -> LibreNMS	Optional
Syslog	UDP/514	Device -> LibreNMS	Optional
HTTPS	TCP/443	User/Agent -> reverse proxy	Productie

Recomandari:

- SNMP trebuie permis numai din IP-ul LibreNMS.
- MySQL, MariaDB si Redis nu se publica in exterior.
- Porturile 8081/8000 nu se expun pe Internet.
- Foloseste un VLAN de management si routing stabil catre echipamente.

6 5. Structura repository

```
GLPI-LibreNMS/
   README.md
   SECURITY.md
   LICENSE
   compose.yml
   .env.example
   docs/
       INSTALL.md
       ARCHITECTURE.md
       CISCO-SNMPV3.md
       MIKROTIK-SNMPV3.md
       GLPI-AGENT.md
       BACKUP-RESTORE.md
       TROUBLESHOOTING.md
       COMPLETE-GUIDE-RO.md
   config/
       cisco/snmpv3-example.txt
       mikrotik/snmpv3-example.rsc
   scripts/
       install-docker.sh
       validate.sh
       deploy.sh
       status.sh
       backup.sh
   .github/workflows/
```

7 6. Instalarea hostului si Docker

7.1 6.1 Pregatirea Ubuntu

```
sudo apt update
sudo apt full-upgrade -y
```

```
sudo reboot
```

Configureaza IP static, DNS, NTP si hostname/FQDN inainte de deployment.

7.2 6.2 Clonarea repository-ului

```
cd /opt
sudo git clone https://github.com/anatolie-ernu/GLPI-LibreNMS.git
sudo chown -R $USER:$USER /opt/GLPI-LibreNMS
cd /opt/GLPI-LibreNMS
```

7.3 6.3 Instalarea Docker

```
sudo bash scripts/install-docker.sh
```

Verificare:

```
docker --version
docker compose version
systemctl status docker --no-pager
```

Daca utilizatorul a fost adaugat in grupul `docker`, executa logout/login inainte de rularea Docker fara `sudo`.

8 7. Configurarea `.env`

```
cd /opt/GLPI-LibreNMS
cp .env.example .env
chmod 600 .env
nano .env
```

Schimba obligatoriu:

```
GLPI_DB_PASSWORD
LIBRENMS_DB_PASSWORD
PUID
PGID
```

Genereaza parole puternice:

```
openssl rand -base64 32
```

Verifica UID/GID:

```
id
```

Nu comite niciodata `.env` in Git. Repository-ul include `.gitignore` si verificare CI, dar acestea nu inlocuiesc secret management-ul operational.

9 8. Imagini si servicii Docker

Configuratia curenta din `.env.example` foloseste:

Serviciu	Imagine
GLPI	`glpi/glpi:11.0.8`
GLPI DB	`mysql:8.4`
LibreNMS	`librenms/librenms:latest`
LibreNMS DB	`mariadb:10`
Redis	`redis:7.2-alpine`

Volume persistente:

- `glpi_db` - MySQL GLPI;
- `glpi_data` - `/var/glpi`;
- `librenms_db` - MariaDB LibreNMS;
- `librenms_data` - `/data`, inclusiv RRD/config/history.

  Pentru productie controlata este recomandat sa testezi toate upgrade-urile in staging. Poti pin-ui imaginile la tag sau digest dupa validare.

10 9. Validare si deployment

10.1 9.1 Validare

```
bash scripts/validate.sh
```

Scriptul verifica:

- existenta `.env`;
- eliminarea placeholder-elor `CHANGE_ME`;
- `docker compose config`;
- sintaxa Bash a scripturilor.

10.2 9.2 Deployment

```
bash scripts/deploy.sh
```

Verificare:

```
docker compose ps
bash scripts/status.sh
```

Loguri:

```
docker compose logs --tail=200 glpi
docker compose logs --tail=200 glpi-db
docker compose logs --tail=200 librenms
docker compose logs --tail=200 librenms-dispatcher
docker compose logs --tail=200 librenms-db
```

Acces initial:

- GLPI: `http://SERVER-IP:8081`
- LibreNMS: `http://SERVER-IP:8000`

11 10. Configurarea initiala GLPI

1. Acceseaza GLPI si finalizeaza wizard-ul initial, daca apare.
2. Configureaza contul administrativ si elimina credentialele implicite.
3. Configureaza entitati, locatii si naming conventions.
4. Verifica automatic actions si cron worker-ul.
5. Configureaza GLPI Agent.
6. Configureaza LDAP/AD doar dupa validarea inventarului de baza.
7. Activeaza HTTPS inainte de rollout larg al agentilor.

GLPI trebuie folosit pentru asset inventory/CMDB, nu drept inlocuitor pentru FDB/LLDP polling in timp real.

12 11. GLPI Agent

Endpoint implicit pentru GLPI 10+ / 11:

`http://<GLPI_SERVER>:8081/front/inventory.php`

Dupa reverse proxy foloseste URL-ul HTTPS canonic.

12.1 11.1 Pilot

1. Instaleaza GLPI Agent pe 2-5 statii reprezentative.
2. Configureaza target-ul catre GLPI.
3. Forteaza inventory.
4. Confirma ca fiecare device apare o singura data.
5. Verifica serial/model/OS/software/IP/MAC.
6. Cauta acelasi MAC in LibreNMS.
7. Valideaza switch/port/VLAN.

12.2 11.2 Rollout GPO

Model recomandat:

```
Computer Configuration
  -> Policies
     -> Software Settings / Startup Script
        -> GLPI Agent MSI installation
```

Exemplu share:

```
\\domain.example\NETLOGON\software\glpi-agent\
```

Rollout rings:

Ring	Populatie
0	IT test devices
1	Departamentul IT
2	Business users selectati
3	Toate statiile administrate
4	Servere, dupa validare separata

Deployment-ul trebuie sa fie idempotent. Nu incorpora credentiale administrative in startup scripts.

13 12. Configurarea initiala LibreNMS

1. Acceseaza `http://SERVER-IP:8000`.
2. Creeaza utilizatorul administrator.
3. Ruleaza validarea:

```
docker compose exec librenms lnms validate
```

4. Adauga un singur Cisco pilot.
5. Verifica interfaces, VLAN, FDB/MAC, LLDP/CDP, sensors si graphs.
6. Adauga un MikroTik pilot.
7. Extinde onboarding-ul numai dupa validarea pilotului.

Un device poate fi adaugat si din CLI:

```
docker compose exec librenms lnms device:add <HOSTNAME_OR_IP>
```

14 13. Cisco IOS / IOS XE - SNMPv3

Standard: **SNMPv3 authPriv**, read-only, ACL catre IP-ul LibreNMS.

```
conf t

ip access-list standard ACL-LIBRENMS-SNMP
 permit host <LIBRENMS_IP>
 deny any log
```

```
exit

snmp-server view LIBRENMS-VIEW iso included
snmp-server group LIBRENMS-GROUP v3 priv read LIBRENMS-VIEW access ACL-LIBRENMS-SNMP
snmp-server user librenms LIBRENMS-GROUP v3 auth sha <AUTH_PASSWORD> priv aes 128 <PRIV_PASSWORD>

snmp-server contact <CONTACT>
snmp-server location <LOCATION>

lldp run
cdp run

end
write memory
```

Foloseste parole diferite pentru autentificare si privacy.

Daca setezi manual SNMP engine-id, configureaza-l inainte de crearea userilor SNMPv3.

Test de pe hostul LibreNMS:

```
sudo apt install -y snmp
snmpwalk -v3 \
  -l authPriv \
  -u librenms \
  -a SHA \
  -A '<AUTH_PASSWORD>' \
  -x AES \
  -X '<PRIV_PASSWORD>' \
  <SWITCH_IP> \
  1.3.6.1.2.1.1
```

Verificari Cisco:

```
show snmp user
show snmp group
show access-lists ACL-LIBRENMS-SNMP
show lldp neighbors detail
show cdp neighbors detail
show mac address-table
```

Dupa polling confirma:

- ports corecte;
- VLAN-uri;
- FDB/MAC;
- LLDP/CDP neighbors;
- diferentierea uplink vs edge/access.

15 14. MikroTik RouterOS - SNMPv3

Configuratia de baza este read-only, cu acces limitat la LibreNMS `/32`.

```
/snmp community
set [find default=yes] read-access=no write-access=no
add name="librenms" \
    addresses=<LIBRENMS_IP>/32 \
    security=private \
    authentication-protocol=SHA1 \
    authentication-password="<AUTH_PASSWORD>" \
    encryption-protocol=AES \
```

```
    encryption-password="<PRIV_PASSWORD>" \
    read-access=yes \
    write-access=no

/snmp
set enabled=yes contact="<CONTACT>" location="<LOCATION>"
```

Optional, daca raspunsul pleaca cu source address gresit:

```
/snmp set src-address=<MANAGEMENT_IP>
```

Verificari:

```
/snmp print detail
/snmp community print detail
/interface bridge host print
/ip neighbor print detail
```

Test SNMP din hostul LibreNMS este identic cu Cisco, folosind IP-ul MikroTik.

15.1 14.1 Enhanced VLAN discovery

LibreNMS poate utiliza helper-ul RouterOS `LNMS_vlans` pentru detalii VLAN. Acesta poate necesita SNMP write pentru a declansa un script RouterOS.

Repository-ul lasa write-access dezactivat implicit. Daca functia este necesara:

1. Restrictioneaza sursa la LibreNMS `/32`.
2. Pastreaza SNMPv3 authPriv.
3. Review-uieste scriptul.
4. Ofera numai policy-urile necesare.
5. Documenteaza exceptia ca acces privilegiat.
6. Monitorizeaza configuration changes.

16 15. Ordinea recomandata de onboarding

1. Un Cisco access switch.
2. Validare SNMPv3, interfaces, VLAN, FDB/MAC, LLDP/CDP.
3. Un MikroTik.
4. Validare SNMPv3, bridge/FDB si neighbors.
5. Core/distribution switches.
6. Restul access switches.
7. GLPI Agent pe cateva endpoint-uri.
8. Corelare end-to-end pentru 5-10 endpoint-uri.
9. Rollout larg al GLPI Agent.
10. Optional traps/syslog dupa stabilizarea polling-ului.

17 16. Corelarea PC -> IP -> MAC -> VLAN -> Switch -> Port

Exemplu:

```
GLPI
  Hostname: HQ-PC-043
  IP:       10.129.21.43
  MAC:      00:25:90:AB:CD:EF

LibreNMS
  cauta MAC 00:25:90:AB:CD:EF
      |
      +-- access port -> endpoint probabil gasit
```

```
      |
      +-- uplink/trunk -> urmeaza LLDP/CDP catre downstream switch
                            |
                            +-- repeta FDB lookup

Rezultat:
  Switch: HQ-ACCESS-SW-04
  Port:   Gi1/0/31
  VLAN:   129
```

Reguli:

- confirma ca FDB entry este recenta;
- confirma VLAN si port state;
- daca portul este trunk, continua downstream;
- coreleaza GLPI Agent IP/MAC cu ARP/FDB;
- tine cont de AP-uri, telefoane IP, mini-switch-uri si bridge-uri.

18 17. Securitate si hardening

- SNMPv3 `authPriv` pentru Cisco si MikroTik.
- SNMP read-only implicit.
- ACL/restrictie `/32` la LibreNMS.
- `.env` cu `chmod 600`, niciodata in Git.
- Databases si Redis fara porturi publice.
- GLPI si LibreNMS in management LAN.
- HTTPS pentru productie.
- MikroTik SNMP write dezactivat implicit.
- Backup separat si restore tests.
- Upgrade numai dupa staging validation.

Nu folosi community `public` sau SNMPv2c in productie daca echipamentul suporta SNMPv3.

19 18. Reverse proxy si HTTPS

In productie foloseste FQDN-uri interne, de exemplu:

Serviciu	FQDN	Backend
GLPI	`glpi.example.local`	`http://docker-host:8081`
LibreNMS	`librenms.example.local`	`http://docker-host:8000`

Recomandari:

- certificat emis de CA interna sau ACME;
- preserve `Host` si `X-Forwarded-*` headers;
- evita SSL inspection care rupe certificatul pentru GLPI Agent;
- activeaza HSTS numai dupa validarea completa HTTPS;
- actualizeaza target-ul GLPI Agent dupa migrare.

20 19. Backup

```
bash scripts/backup.sh
```

Backup-ul include:

- GLPI database dump;
- LibreNMS database dump;

- `/var/glpi` persistent data;
- `/data` LibreNMS persistent data;
- `compose.yml`;
- template `.env`;
- `SHA256SUMS`.

Fisierul real `.env` nu este copiat intentionat.

Verificare:

```
cd <BACKUP_DIRECTORY>
sha256sum -c SHA256SUMS
```

21 20. Restore

Ordine:

1. Opreste serviciile aplicative.
2. Restaureaza datele persistente.
3. Restaureaza bazele de date.
4. Porneste stack-ul.
5. Ruleaza `lnms validate`.
6. Verifica un endpoint GLPI si un switch.

Exemplu:

```
BACKUP=/opt/backups/glpi-librenms/20260914_150000

docker compose stop glpi librenms librenms-dispatcher

gzip -dc "$BACKUP/glpi-data.tar.gz" | \
  docker compose run --rm -T glpi sh -lc 'rm -rf /var/glpi/* && tar -C /var/glpi -xzf -'

gzip -dc "$BACKUP/librenms-data.tar.gz" | \
  docker compose run --rm -T librenms sh -lc 'rm -rf /data/* && tar -C /data -xzf -'

gzip -dc "$BACKUP/glpi-db.sql.gz" | \
  docker compose exec -T glpi-db sh -lc \
  'mysql -u"$MYSQL_USER" -p"$MYSQL_PASSWORD" "$MYSQL_DATABASE"'

gzip -dc "$BACKUP/librenms-db.sql.gz" | \
  docker compose exec -T librenms-db sh -lc \
  'mariadb -u"$MYSQL_USER" -p"$MYSQL_PASSWORD" "$MYSQL_DATABASE"'

docker compose up -d
bash scripts/status.sh
docker compose exec librenms lnms validate
```

> Un backup care nu a fost restaurat niciodata nu este demonstrat. Executa periodic restore tests si inregistreaza RPO/RTO.

22 21. Upgrade si mentenanta

Flux recomandat:

1. Citeste release notes.
2. Executa backup.
3. Verifica checksums.
4. Testeaza in staging.
5. Actualizeaza tag/digest numai dupa validare.

6. Ruleaza pull/up.
7. Verifica status si `lnms validate`.
8. Testeaza login, inventory si polling.
9. Pastreaza rollback plan.

```
bash scripts/backup.sh
docker compose pull
docker compose up -d
bash scripts/status.sh
docker compose exec librenms lnms validate
```

23 22. Troubleshooting

23.1 22.1 Status si logs

```
docker compose ps
bash scripts/status.sh

docker compose logs --tail=200 glpi
docker compose logs --tail=200 glpi-db
docker compose logs --tail=200 librenms
docker compose logs --tail=200 librenms-dispatcher
docker compose logs --tail=200 librenms-db
```

23.2 22.2 SNMPv3 nu functioneaza

Verifica:

- routing catre management IP;
- UDP/161;
- username;
- `authPriv`;
- auth algorithm/password;
- privacy algorithm/password;
- ACL Cisco sau MikroTik `/32`;
- source IP real al LibreNMS;
- VRF/routing table folosit de device.

```
snmpwalk -v3 \
  -l authPriv \
  -u librenms \
  -a SHA \
  -A '<AUTH_PASSWORD>' \
  -x AES \
  -X '<PRIV_PASSWORD>' \
  <DEVICE_IP> \
  1.3.6.1.2.1.1
```

23.3 22.3 MAC apare pe uplink

Este normal. Urmareste LLDP/CDP/neighbor topology downstream si repeta FDB lookup pana la portul edge final.

23.4 22.4 Duplicate GLPI computers

Verifica:

- agent identity;
- serial number quality;
- Windows cloned images;
- assets create manual;

- reinstall/enrollment history.

Nu sterge duplicatele pana nu stabilesti recordul autoritativ.

23.5 22.5 Disk usage / restart loop

```
df -h
docker system df
docker volume ls
docker inspect <container>
docker compose logs --tail=300 <service>
```

24 23. Runbook operational

Frecventa	Actiune
Zilnic	Alerte LibreNMS, restart/unhealthy, disk critic
Saptamanal	Device down, polling errors, GLPI agent failures, backup status
Lunar	Patching host, image review, capacity trend, access review
Trimestrial	Restore test, SNMP ACL/users, certificate expiry
Schimbare majora	Backup, staging, change ticket, rollback plan

Quick health check:

```
cd /opt/GLPI-LibreNMS
bash scripts/status.sh
docker compose exec librenms lnms validate
df -h
docker system df
```

25 24. Checklist productie

- ☐ Ubuntu 24.04 patch-uit, IP static, DNS si NTP OK.
- ☐ Docker Engine si Compose plugin instalate.
- ☐ `.env` cu parole puternice si `chmod 600`.
- ☐ GLPI si LibreNMS fara restart loop.
- ☐ GLPI functional si cron worker activ.
- ☐ LibreNMS `lnms validate` fara FAIL critic.
- ☐ Cisco pilot SNMPv3 authPriv + ACL + LLDP/CDP validat.
- ☐ MikroTik pilot SNMPv3 + `/32` + write disabled validat.
- ☐ FDB/MAC si VLAN validate.
- ☐ GLPI Agent pilot pe 2-5 endpoint-uri.
- ☐ Corelare GLPI MAC -> LibreNMS switch/port confirmata.
- ☐ Reverse proxy + HTTPS configurat.
- ☐ Backup automatizat.
- ☐ Restore test executat.
- ☐ DB/Redis nu sunt expuse extern.
- ☐ CI repository green.

26 Anexa A - `.env.example`

```
# General
TZ=Europe/Chisinau
PUID=1000
```

```
PGID=1000

# Published ports
GLPI_HTTP_PORT=8081
LIBRENMS_HTTP_PORT=8000

# Images
GLPI_IMAGE=glpi/glpi:11.0.8
GLPI_DB_IMAGE=mysql:8.4
LIBRENMS_IMAGE=librenms/librenms:latest
LIBRENMS_DB_IMAGE=mariadb:10
REDIS_IMAGE=redis:7.2-alpine

# GLPI database
GLPI_DB_NAME=glpi
GLPI_DB_USER=glpi
GLPI_DB_PASSWORD=CHANGE_ME_STRONG_GLPI_DB_PASSWORD
GLPI_CRONTAB_ENABLED=1
GLPI_SKIP_AUTOINSTALL=false
GLPI_SKIP_AUTOUPDATE=false

# LibreNMS database
LIBRENMS_DB_NAME=librenms
LIBRENMS_DB_USER=librenms
LIBRENMS_DB_PASSWORD=CHANGE_ME_STRONG_LIBRENMS_DB_PASSWORD

# LibreNMS PHP/runtime settings
LIBRENMS_MEMORY_LIMIT=512M
LIBRENMS_MAX_INPUT_VARS=2000
LIBRENMS_UPLOAD_MAX_SIZE=32M
LIBRENMS_OPCACHE_MEM_SIZE=256

# Deployment metadata
BACKUP_DIR=/opt/backups/glpi-librenms
```

27 Anexa B - Configuratie Cisco

Fisier repository: `config/cisco/snmpv3-example.txt`.

```
conf t
ip access-list standard ACL-LIBRENMS-SNMP
 permit host <LIBRENMS_IP>
 deny any log
exit
snmp-server view LIBRENMS-VIEW iso included
snmp-server group LIBRENMS-GROUP v3 priv read LIBRENMS-VIEW access ACL-LIBRENMS-SNMP
snmp-server user librenms LIBRENMS-GROUP v3 auth sha <AUTH_PASSWORD> priv aes 128 <PRIV_PASSWORD>
snmp-server contact <CONTACT>
snmp-server location <LOCATION>
lldp run
cdp run
end
write memory
```

28 Anexa C - Configuratie MikroTik

Fisier repository: `config/mikrotik/snmpv3-example.rsc`.

```
/snmp community
set [find default=yes] read-access=no write-access=no
add name="librenms" addresses=<LIBRENMS_IP>/32 security=private authentication-protocol=SHA1 authentication-
/snmp
set enabled=yes contact="<CONTACT>" location="<LOCATION>"
# /snmp set src-address=<MANAGEMENT_IP>
/snmp print detail
/snmp community print detail
```

29 Anexa D - Comenzi utile

Comanda	Scop
`sudo bash scripts/install-docker.sh`	Instaleaza Docker
`bash scripts/validate.sh`	Valideaza configuratia
`bash scripts/deploy.sh`	Deploy stack
`bash scripts/status.sh`	Status + HTTP checks
`bash scripts/backup.sh`	Backup complet
`docker compose exec librenms lnms validate`	Validate LibreNMS
`docker compose logs --tail=200 <service>`	Troubleshooting

30 Anexa E - Referinte oficiale

- Repository: https://github.com/anatolie-ernu/GLPI-LibreNMS
- GLPI Docker: https://github.com/glpi-project/docker-images
- GLPI documentation: https://help.glpi-project.org/documentation/
- GLPI Agent: https://glpi-agent.readthedocs.io/
- LibreNMS Docker: https://docs.librenms.org/Installation/Docker/
- LibreNMS documentation: https://docs.librenms.org/
- Docker Engine Ubuntu: https://docs.docker.com/engine/install/ubuntu/
- MikroTik RouterOS documentation: https://manual.mikrotik.com/
- Cisco documentation: https://www.cisco.com/

31 Anexa F - Glosar

Termen	Explicatie
CMDB	Configuration Management Database
FDB	Forwarding Database / tabela MAC
ARP	Mapare IPv4 -> MAC
LLDP	Link Layer Discovery Protocol
CDP	Cisco Discovery Protocol
SNMPv3 authPriv	SNMP cu autentificare si criptare
RRD	Round Robin Database
Access port	Port edge asociat endpoint-ului
Trunk/uplink	Port intre switch-uri / multiple VLAN-uri
RPO	Recovery Point Objective
RTO	Recovery Time Objective

32 Politica Service Desk P1-P4, SLA si escaladare

Repository-ul defineste acum o politica explicita pentru GLPI Service Desk, versionata in Git.

Matricea de baza este:

Impact Urgency	High	Medium	Low
High	P1 Critical	P2 High	P3 Medium
Medium	P2 High	P3 Medium	P4 Low
Low	P3 Medium	P4 Low	P4 Low

SLA baseline:

Prioritate	TTO	TTR
P1 Critical	15 min	2 h
P2 High	30 min	4 h
P3 Medium	4 h	1 business day
P4 Low	1 business day	3 business days

Model suport:

```
L0  Self-Service / Knowledge Base
L1  SD-L1-ServiceDesk
L2  Network / Infrastructure / Security / Applications / Business Systems
L3  Senior / IT Management / Vendor Coordination
```

Politica completa, regulile de escaladare si fisierele source-of-truth sunt in:

- `docs/GLPI-PRIORITY-SLA-ESCALATION-MATRIX.md`
- `config/glpi/servicedesk-policy.yaml`
- `config/glpi/*.csv`

Configurarea efectiva in GLPI trebuie validata prin teste accelerate de SLA si prin rapoarte de conformitate inainte de go-live.